

Bayesian Logistic Regression for Heart Disease Prediction

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Abstract—Heart disease remains one of the leading causes of mortality worldwide, highlighting the importance of accurate and reliable risk prediction to support clinical decision making. Although the Heart Disease UCI dataset has been widely studied, systematic comparisons between frequentist logistic regression and Bayesian logistic regression with different prior specifications remain limited. This study aims to compare the predictive performance of a frequentist logistic regression model with two Bayesian logistic regression models employing different normal prior variances. The analysis is conducted using the Heart Disease UCI dataset, and Bayesian inference is performed via Markov Chain Monte Carlo (MCMC). Model performance is evaluated using predictive metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC, as well as Bayesian model comparison criteria such as DIC, LOOIC, and WAIC. The results indicate that the Bayesian logistic regression model with a Normal(0, 1) prior achieves the best overall predictive performance, exhibiting higher accuracy, precision, and F1-score compared to the alternative Bayesian model and the frequentist approach. While the frequentist model attains a slightly higher ROC-AUC, the difference is marginal and does not outweigh the overall balanced performance of the Bayesian model.

Keywords—Bayesian Logistic Regression, Frequentist Logistic Regression, Heart Disease Prediction, MCMC

I. INTRODUCTION

A. Background

Heart disease remains one of the leading causes of mortality worldwide, making early and accurate risk prediction essential for supporting clinical decision making. The Heart Disease UCI dataset is widely used as a benchmark due to its comprehensive clinical attributes, enabling consistent evaluation of predictive modeling techniques. While numerous

studies have applied frequentist or machine learning approaches to this dataset, systematic comparisons between Bayesian and frequentist frameworks are still limited.

Bayesian logistic regression provides advantages such as incorporating prior knowledge and explicitly quantifying uncertainty. However, the performance of Bayesian models can vary depending on the specification of the prior. In contrast, frequentist logistic regression relies solely on observed data without prior information. These differences motivate the investigation of how varying prior specifications in Bayesian models affect predictive performance compared to the frequentist approach.

This research is motivated by the need to provide empirical insights into the strengths and limitations of frequentist logistic regression and Bayesian logistic regression with different priors, supporting better methodological choices for future medical prediction tasks.

B. Problem Statement

Despite widespread use of the Heart Disease UCI dataset, there is a lack of systematic analysis comparing frequentist logistic regression with Bayesian logistic regression models using different prior specifications. It remains unclear how prior specification influences predictive accuracy, parameter stability, and uncertainty estimation in Bayesian inference. Additionally, it is not yet established whether Bayesian models with different priors provide measurable advantages over frequentist models that rely solely on observed data. This gap forms the core problem addressed in this study.

C. Objectives

The objectives of this study are:

1. Compare the predictive performance of frequentist logistic regression and Bayesian logistic regression models with different priors using the Heart Disease UCI dataset.
2. Evaluate the impact of prior selection on accuracy.
3. Identify the most effective modeling approach for heart disease risk prediction based on empirical performance metrics.

II. LITERATURE REVIEW

A. Existing Studies

Existing research on heart disease prediction using the Heart Disease UCI dataset primarily focuses on traditional statistical models and machine-learning classifiers. Frequentist methods such as logistic regression, support vector machines, k-nearest neighbors, decision trees, and ensemble techniques have been widely applied, consistently demonstrating moderate to high predictive accuracy [1,8]. Several studies also utilize neural networks and hybrid optimization techniques to improve performance [2,5]. However, despite the extensive exploration of algorithmic approaches, most studies center on point estimation and deterministic predictions, offering limited attention to uncertainty quantification and the incorporation of prior clinical knowledge.

B. Bayesian Analysis in Predictive Modeling

For the past few years, Bayesian application has increased drastically, providing solutions for problems that a traditional approach fails to deal with [4]. The Bayesian Logistic Regression Model demonstrates outstanding performance across metrics, namely accuracy, precision, recall, F1-Score, and AUC. This indicates Bayesian's outstanding ability to produce accurate predictions [7].

C. Justify why the Bayesian approach

Bayesian method offers an explanatory advantage in predictive modeling, especially in healthcare, because it does not only provide outcome prediction but also quantify the uncertainty surrounding those predictions. In the medical field, ambiguity is common and incorrect decisions can be costly. Hence, decision making should be taken into account. With the help of Bayesian models, we can get high prediction accuracy and interpretable confidence measures. These characteristics make the Bayesian approach suitable for medical purposes that require reliability to ensure safer and more personalized patient care [6]. Bayesian logistic regression demonstrates greater robustness than classical maximum likelihood because it produces more stable parameter estimates even under challenging data conditions. Bayesian variants yielded more consistent coefficients than MLE as one of the frequentist approaches. This finding indicates Bayesian's resistance to unstable data conditions. This stability and reliability make Bayesian logistic regression

a stronger choice when model precision and consistency are critical [9].

III. METHODOLOGY

A. Data Description

The dataset used in this study is the Heart Disease Dataset [11], obtained from Kaggle. It contains 303 patient records, each providing clinical and physiological measurements relevant to cardiovascular health. The main purpose of using this dataset is to analyze how various medical covariates influence the occurrence of heart disease and to compare the performance of several predictive modeling approaches.

The dataset includes the following covariates :

- Age : The patient's age in years.
- Sex : Gender (1 = male, 0 = female).
- Chest Pain Type (cp) : Types of chest pain experienced by the patient (0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic).
- Resting Blood Pressure (trestbps) : Resting blood pressure measured in mm Hg.
- Cholesterol (chol) : Cholesterol level in mg/dL.
- Fasting Blood Sugar (fbs) : Indicates whether fasting blood sugar is > 120 mg/dL (1 = yes, 0 = no).
- Resting ECG Results (restecg) : Resting electrocardiogram results (0 = normal, 1 = ST-T wave abnormality, 2 = showing left ventricular hypertrophy).
- Maximum Heart Rate Achieved (thalach) : The maximum heart rate achieved during a stress test.
- Exercise-Induced Angina (exang) : Indicates whether the patient experienced angina during exercise (1 = yes, 0 = no).
- ST Depression Induced by Exercise (oldpeak) : ST segment depression included by exercise relative to rest.
- Slope of the Peak Exercise ST Segment (slope) : Slope category of the peak exercise ST segment (0, 1, or 2).
- Number of Major Vessels (ca) : Number of major vessels (0-3) observed via fluoroscopy.
- Thalassemia (thal) : Thalassemia status (e.g., normal, fixed defect, reversible defect).
- Target : Indicator of heart disease (0 = no heart disease, 1 = have heart disease).

B. Bayesian Logistic Regression

Logistic regression is a statistical method used to model the relationship between a response variable and one or more predictor variables, where the response variable is binary. A binary variable has two possible outcomes, typically coded as 0 (failure) and 1 (success). Logistic regression can be applied for prediction or classification tasks and can also identify significant factors that influence the response variable.

Furthermore, logistic regression is flexible in handling datasets where the response variable is categorical and the predictor variables can be either categorical or continuous [3].

C. Prior Distribution, Likelihood, Posterior

Binary Outcome and Logit Model

Let (x_i, y_i) denote the i -th observation, where:

- $y_i \in (0,1)$ is a binary outcome variable,
- $x_i = (x_{i1}, x_{i2}, x_{i3}, \dots, x_{ip})^T$ is a vector of predictor variables,
- $\beta = (\beta_1, \beta_2, \beta_3, \dots, \beta_p)^T$ is a vector of regression coefficients.

The logistic regression model assumes that the **log-odds (logit)** of the probability of success is a linear function of the predictors:

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \beta,$$

where:

$$p_i = P(y_i = 1 \mid x_i, \beta)$$

exponentiating both sides yields:

$$\begin{aligned} \frac{p_i}{1-p_i} &= \exp(x_i^T \beta) \\ p_i &= \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} = \frac{1}{1 + \exp(-x_i^T \beta)} \end{aligned}$$

Prior Distribution

In the Bayesian framework, prior beliefs about the regression coefficients are incorporated through a prior distribution. A multivariate Gaussian prior is commonly assumed:

$$\beta \sim \mathcal{N}(0, \tau^2 I),$$

where τ^2 represents the prior variance and I is the identity matrix.

The corresponding prior density function is:

$$p(\beta) = (2\pi\tau^2)^{-p/2} \exp\left(-\frac{1}{2\tau^2} \beta^T \beta\right)$$

This prior imposes shrinkage on the coefficients and acts as an ℓ_2 -regularization term.

In this study, two different multivariate Gaussian priors are considered to assess the sensitivity of the Bayesian logistic regression model to the choice of prior variance.

- The first prior is specified as a normal distribution with **mean 0** and **variance 2**, which provides relatively weaker regularization and allows greater flexibility in the estimated coefficients.
- The second prior is defined as a normal distribution with **mean 0** and **variance 1**, imposing stronger shrinkage on the coefficients and encouraging more conservative parameter estimates.

Likelihood

Since the response variable y_i is binary, it is assumed to follow a Bernoulli distribution:

$$y_i \mid p_i \sim \text{Bernoulli}(p_i)$$

$$P(y_i \mid p_i) = p_i^{y_i} (1 - p_i)^{1-y_i}$$

Assuming conditional independence of observations, the likelihood function for the full dataset is:

$$\begin{aligned} L(\beta) &= \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i} \\ L(\beta) &= \prod_{i=1}^n \left[\frac{1}{1 + \exp(-x_i^T \beta)} \right]^{y_i} \left[\frac{\exp(-x_i^T \beta)}{1 + \exp(-x_i^T \beta)} \right]^{1-y_i} \end{aligned}$$

Posterior

Using Bayes' theorem, the posterior distribution of the parameters is given by:

$$p(\beta \mid X, y) \propto p(y \mid X, \beta) p(\beta)$$

$$p(\beta \mid X, y) \propto \left[\prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i} \right] \exp\left(-\frac{1}{2\tau^2} \beta^T \beta\right)$$

$$p(\beta \mid X, y) \propto \prod_{i=1}^n \left[\frac{1}{1 + \exp(-x_i^T \beta)} \right]^{y_i} \left[\frac{\exp(-x_i^T \beta)}{1 + \exp(-x_i^T \beta)} \right]^{1-y_i} \exp\left(-\frac{1}{2\tau^2} \beta^T \beta\right)$$

$$\log p(\beta \mid X, y) = \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)] - \frac{1}{2\tau^2} \beta^T \beta + C$$

D. Markov Chain Monte Carlo

Markov Chain Monte Carlo (MCMC) is a computational sampling technique used to characterize a probability distribution without requiring complete knowledge of its analytical properties. This method relies on randomly generating samples from the target distribution, provided that the probability density can be evaluated for given values. One of the main advantages of MCMC is its ability to handle complex distributions whose mathematical forms are difficult to analyze directly. The term MCMC reflects two fundamental concepts, Monte Carlo methods and Markov chains. Monte Carlo methods refer to statistical techniques that estimate the characteristics of a distribution by analyzing randomly generated samples. For instance, instead of deriving the mean of a normal distribution analytically, a Monte Carlo approach estimates the mean by computing the average of a large number of random samples drawn from the distribution. This approach is particularly advantageous when sampling is straightforward while analytical calculations are cumbersome. The Markov chain component of MCMC describes the sequential mechanism by which samples are generated. Each sample is produced based solely on the immediately preceding sample, forming a chain of dependent observations. Although successive samples are correlated, the defining property of a Markov chain is that each new sample depends only on the previous one and not on earlier samples. This property enables the sampling process to efficiently explore the target distribution [10].

The MCMC sampling procedure is configured using the following hyperparameters:

- Chains (**chains = 3**): Three independent Markov chains are run in parallel to improve the reliability of posterior estimates and to facilitate convergence diagnostics across chains.
- Burn-in (**burn = 10000**): The initial 10000 iterations of each chain are discarded to reduce the influence of initial values and ensure that the samples are drawn from the stationary distribution.
- Iterations (**iter = 50000**): Each chain is run for a total of 50000 iterations to generate a sufficiently large sample for stable posterior inference and accurate estimation of posterior summaries.

IV. RESULT & DISCUSSION

A. Model Estimation Results

1. Bayesian Logistic Regression Model 1

This model used weakly informative normal priors with mean zero, variance 2, and precision 0.5 [Normal(0, 0.5)] for all regression coefficients.

```
Iterations = 11001:61000
Thinning interval = 1
Number of chains = 3
Sample size per chain = 50000
```

1. Empirical mean and standard deviation for each variable, plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
beta[1]	1.6863	0.7913	0.0020430	0.012477
beta[2]	0.1266	0.3214	0.0008298	0.001423
beta[3]	-1.5517	0.4789	0.0012365	0.004113
beta[4]	0.8081	0.2031	0.0005243	0.001324
beta[5]	-0.3829	0.2436	0.0006289	0.000962
beta[6]	-0.2464	0.2967	0.0007662	0.001370
beta[7]	-0.1835	0.5383	0.0013899	0.002691
beta[8]	0.6008	0.3757	0.0009700	0.002058
beta[9]	0.7623	0.3803	0.0009820	0.001848
beta[10]	-0.8942	0.4284	0.0011061	0.002195
beta[11]	-0.9987	0.4178	0.0010786	0.002274
beta[12]	1.0749	0.3681	0.0009503	0.004321
beta[13]	-0.9272	0.2190	0.0005654	0.001089
beta[14]	-0.8093	0.2835	0.0007319	0.003997

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
beta[1]	0.1394	1.14936	1.6870	2.22237	3.24303
beta[2]	-0.5023	-0.08975	0.1254	0.34284	0.75912
beta[3]	-2.5016	-1.87169	-1.5462	-1.22430	-0.62834
beta[4]	0.4191	0.66978	0.8047	0.94413	1.21446
beta[5]	-0.8647	-0.54518	-0.3810	-0.21781	0.08855
beta[6]	-0.8335	-0.44541	-0.2465	-0.04638	0.33490
beta[7]	-1.2323	-0.54784	-0.1865	0.17999	0.87557
beta[8]	-0.1291	0.34792	0.5987	0.85378	1.33997
beta[9]	0.0334	0.50375	0.7576	1.01367	1.52585
beta[10]	-1.7322	-1.18522	-0.8942	-0.60516	-0.05397
beta[11]	-1.8384	-1.27532	-0.9932	-0.71526	-0.19463
beta[12]	0.3625	0.82574	1.0707	1.32146	1.80426
beta[13]	-1.3723	-1.07106	-0.9224	-0.77771	-0.50861
beta[14]	-1.3696	-0.99920	-0.8092	-0.61957	-0.25235

Fig. 1. Summary of Bayesian Logistic Model 1

Model 1 was estimated using Markov Chain Monte Carlo (MCMC) sampling with three parallel chains. After discarding the burn-in period, a total of 150,000 posterior samples were retained, indicating stable and reliable posterior inference.

Posterior summaries are reported in terms of posterior means, standard deviations, Monte Carlo standard errors, and 95% credible intervals. The coefficient β_1 represents the **intercept**, while the remaining coefficients correspond to the predictor variables. The posterior mean indicates the direction of the effect, with positive values corresponding to an increase and negative values corresponding to a decrease in the log-odds of heart disease. However, inference is primarily based on the 95% credible intervals, where an effect is considered meaningful when the interval does not include **zero**, as this reflects the associated posterior uncertainty.

Based on the posterior credible intervals, several regression coefficients show clear evidence of association with the outcome. Specifically, coefficients β_4 (**cp**), β_9 (**thalach**), and β_{12} (**slope**) have 95% credible intervals entirely above zero, indicating positive effects on the log-odds of the outcome. In contrast, coefficients β_3 (**sex**), β_{10} (**exang**), β_{11} (**oldpeak**), β_{13} (**ca**), and β_{14} (**thal**) have credible intervals entirely below zero, suggesting negative effects on the log-odds of the outcome.

The remaining covariates (β_2 [**age**], β_5 [**trestbps**], β_6 [**chol**], β_7 [**fbs**], and β_8 [**restecg**]) have credible intervals that include zero, indicating that there is insufficient evidence to conclude a clear directional effect for these predictors given the observed data and prior assumptions.

Overall, the results suggest that the outcome is influenced by a subset of predictors with either positive or negative effects, while other predictors show greater posterior uncertainty and do not demonstrate a clear association with the outcome.

2. Bayesian Logistic Regression Model 2

This model also used weakly informative normal priors with mean zero, variance 1, and precision 1 [Normal(0, 1)] for all regression coefficients, implying a smaller prior variance and stronger shrinkage toward zero compared to a precision of 0.5.

```
Iterations = 11001:61000
Thinning interval = 1
Number of chains = 3
Sample size per chain = 50000
```

1. Empirical mean and standard deviation for each variable, plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
beta[1]	1.24719	0.6768	0.0017475	0.0096775
beta[2]	0.09953	0.3030	0.0007823	0.0013016
beta[3]	-1.34209	0.4332	0.0011184	0.0034433
beta[4]	0.78492	0.1939	0.0005006	0.0012098
beta[5]	-0.34995	0.2343	0.0006050	0.0008907
beta[6]	-0.20983	0.2816	0.0007271	0.0012019
beta[7]	-0.13019	0.4908	0.0012673	0.0021383
beta[8]	0.56268	0.3507	0.0009055	0.0018504
beta[9]	0.69153	0.3524	0.0009100	0.0016560
beta[10]	-0.81839	0.4023	0.0010388	0.0020682
beta[11]	-0.93879	0.3902	0.0010075	0.0019168
beta[12]	1.06677	0.3382	0.0008732	0.0037130
beta[13]	-0.88637	0.2088	0.0005390	0.0009841
beta[14]	-0.69794	0.2609	0.0006737	0.0034436

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
beta[1]	-0.07115	0.7877	1.24649	1.7042	2.57080
beta[2]	-0.49452	-0.1052	0.09959	0.3030	0.69201
beta[3]	-2.20565	-1.6313	-1.33550	-1.0465	-0.51226
beta[4]	0.41403	0.6529	0.78189	0.9148	1.17354
beta[5]	-0.81449	-0.5072	-0.34729	-0.1913	0.10605
beta[6]	-0.76392	-0.3991	-0.20979	-0.0192	0.33958
beta[7]	-1.08391	-0.4617	-0.13373	0.2005	0.83810
beta[8]	-0.12000	0.3257	0.56141	0.7973	1.25516
beta[9]	0.01205	0.4529	0.68629	0.9238	1.39760
beta[10]	-1.60714	-1.0892	-0.81617	-0.5471	-0.03068
beta[11]	-1.71078	-1.1990	-0.93501	-0.6721	-0.18869
beta[12]	0.41554	0.8367	1.06352	1.2935	1.73866
beta[13]	-1.30761	-1.0238	-0.88230	-0.7449	-0.48900
beta[14]	-1.21429	-0.8711	-0.69623	-0.5232	-0.18676

Fig. 2. Summary of Bayesian Logistic Model 2

Model 2 was fitted using a Bayesian logistic regression framework implemented via Markov Chain Monte Carlo (MCMC) sampling with three parallel chains. Following the removal of the burn-in iterations, 150,000 posterior draws were retained for inference, suggesting satisfactory convergence and stable posterior estimates.

Parameter estimates are summarized using posterior means, posterior standard deviations, Monte Carlo standard errors, and 95% credible intervals. The posterior mean indicates the direction of the effect, with positive values corresponding to an increase and negative values corresponding to a decrease in the log-odds of the outcome. The **intercept** term is denoted by β_1 , whereas the remaining coefficients correspond to the explanatory variables. Evidence for an association is inferred when the 95% credible interval of a coefficient does not cross zero.

The posterior results indicate that several predictors are credibly associated with the outcome. In particular, coefficients β_4 (**cp**), β_9 (**thalach**), and β_{12} (**slope**) show positive associations, as their credible intervals lie entirely above zero. In contrast, coefficients β_3 (**sex**), β_{10} (**exang**), β_{11} (**oldpeak**), β_{13} (**ca**), and β_{14} (**thal**) exhibit negative associations with the outcome, supported by credible intervals fully below zero. For the remaining predictors (β_2 [**age**], β_5 [**trestbps**], β_6 [**chol**], β_7 [**fbs**], and β_8 [**restecg**]), the credible intervals include zero, indicating considerable posterior uncertainty and a lack of strong evidence for a directional effect.

Taken together, the results from Model 2 indicate that only a subset of predictors exhibits consistent positive or negative effects on the outcome, whereas the remaining variables show no clear evidence of association under the current model assumptions.

3. Frequentist Logistic Regression

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.5864	1.0321	2.506	0.012212 *
age	0.1439	0.3361	0.428	0.668576
sex	-1.7516	0.5362	-3.267	0.001089 **
cp	0.7487	0.2070	3.617	0.000298 ***
trestbps	-0.3917	0.2454	-1.596	0.110422
chol	-0.2843	0.3087	-0.921	0.357100
fbs	-0.3309	0.5882	-0.563	0.573702
restecg	0.5846	0.3918	1.492	0.135634
thalach	0.8267	0.4059	2.037	0.041665 *
exang	-0.9561	0.4514	-2.118	0.034153 *
oldpeak	-1.0152	0.4406	-2.304	0.021197 *
slope	0.8975	0.4033	2.225	0.026053 *
ca	-0.8885	0.2177	-4.082	4.47e-05 ***
thal	-0.9801	0.3209	-3.054	0.002259 **

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 335.05 on 242 degrees of freedom
Residual deviance: 167.14 on 229 degrees of freedom
AIC: 195.14

Based on the model estimation results, the variables found to significantly influence the probability of heart disease are **sex**, **cp**, **thalach**, **exang**, **oldpeak**, **slope**, **ca**, and **thal**. All of these predictors have **p-values below 0.05**, indicating statistically significant effects on the outcome. Among these variables, **cp** and **ca** stand out as the most significant predictors, as they both show the smallest p-values. Positive coefficients, such as those for **cp**, **thalach**, and **slope**, suggest an increased likelihood of heart disease, while negative coefficients for **sex**, **exang**, **oldpeak**, **ca**, and **thal** indicate a reduced likelihood. In terms of model quality, the reduction in deviance from 335.05 to 167.14 demonstrates a substantial improvement in model fit compared to the null model. Additionally, the AIC value of 195.14 reflects a reasonably competitive level of penalized model fit for this dataset.

B. Posterior Distribution

1. Bayesian Logistic Regression Model 1

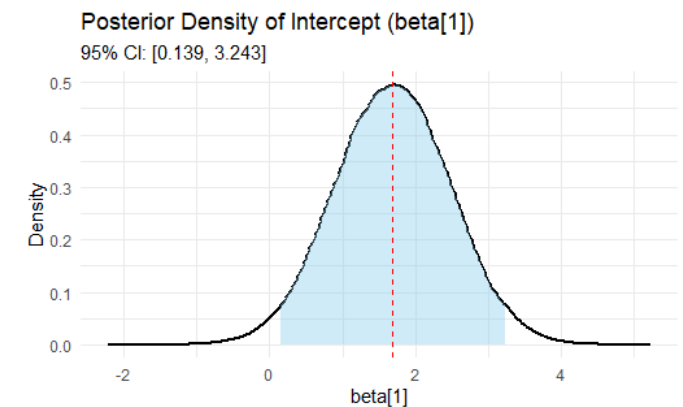


Fig. 3. Posterior Distribution Plot of β_1 in Bayesian Model 1

The posterior distribution of the intercept (β_1) reflects the baseline log-odds of heart disease when all predictor variables are held at their reference levels (categorical variables) or zero (numerical variables). Its credible interval suggests a stable and identifiable baseline probability.

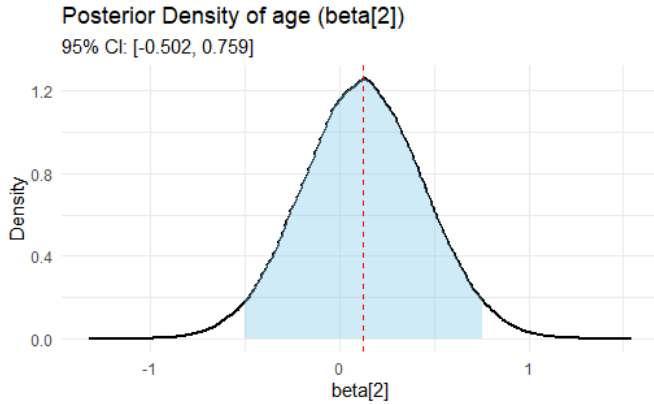


Fig. 4. Posterior Distribution Plot of β_2 in Bayesian Model 1

The posterior distribution of *age* has a 95% credible interval $[-0.502, 0.759]$, which includes zero. This indicates **weak posterior evidence** of a systematic association between *age* and the probability of heart disease.

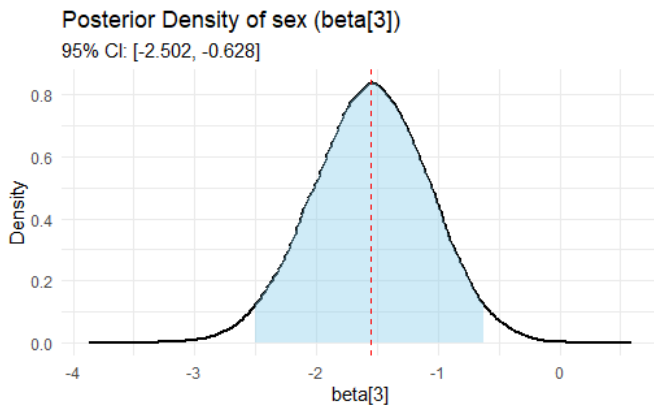


Fig. 5. Posterior Distribution Plot of β_3 in Bayesian Model 1

The posterior distribution of *sex* lies entirely below zero, with a 95% credible interval $[-2.502, -0.628]$. This provides strong posterior evidence of a **negative association** between *sex* and heart disease, suggesting that one sex category is associated with a lower probability of heart disease compared to the reference group.

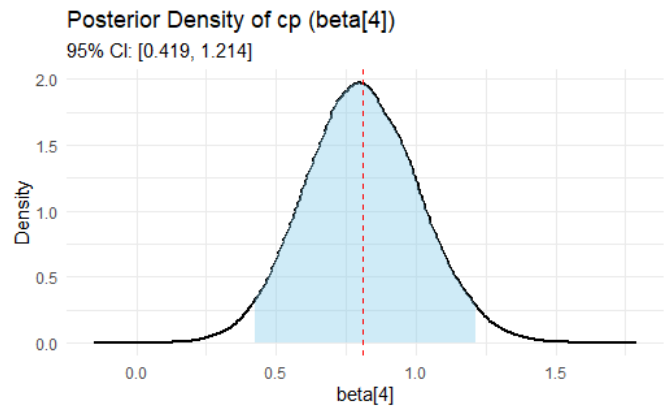


Fig. 6. Posterior Distribution Plot of β_4 in Bayesian Model 1

The posterior distribution of *cp* lies entirely above zero, with a 95% credible interval $[0.419, 1.214]$, indicating strong posterior evidence of a **positive association**. This suggests that chest pain type is an important predictor of heart disease presence.

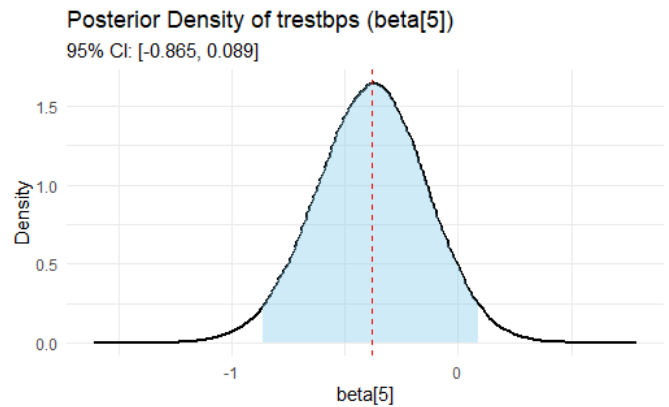


Fig. 7. Posterior Distribution Plot of β_5 in Bayesian Model 1

The 95% credible interval for *trestbps* is $[-0.865, 0.089]$, which includes zero, suggesting **weak posterior evidence** of a consistent association with the probability of heart disease.

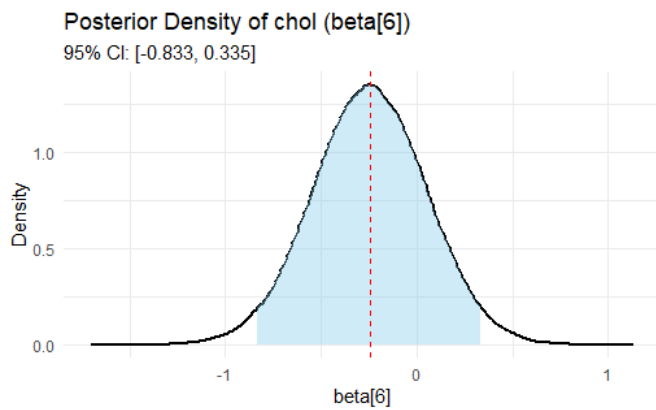


Fig. 8. Posterior Distribution Plot of β_6 in Bayesian Model 1

The posterior distribution of *chol* has a 95% credible interval [-0.833, 0.335], which includes zero. This indicates **weak posterior evidence** of a consistent association between serum cholesterol levels and the probability of heart disease.

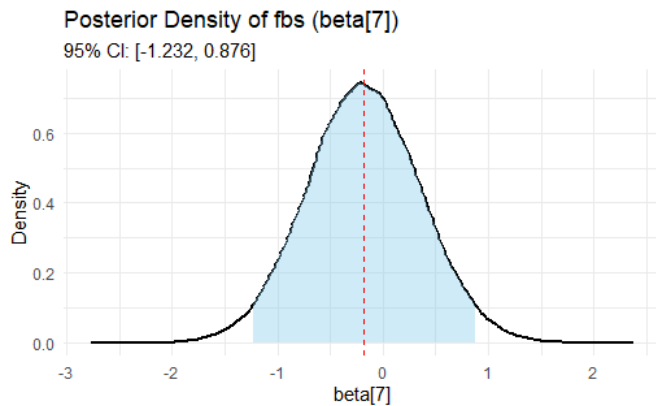


Fig. 9. Posterior Distribution Plot of β_7 in Bayesian Model 1

The posterior distribution of *fbs* has a wide 95% credible interval [-1.232, 0.876], which includes zero, indicating **weak posterior evidence** of an association with heart disease.

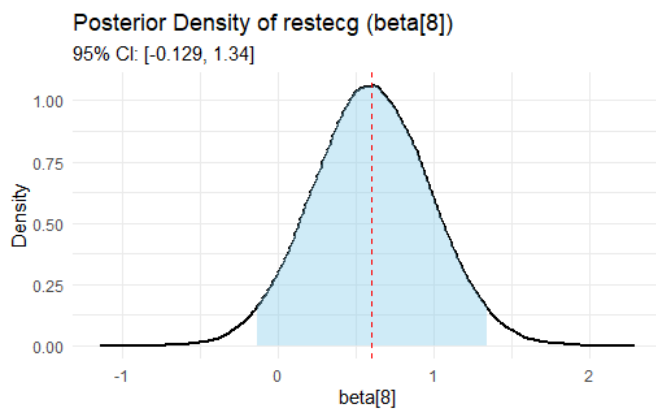


Fig. 10. Posterior Distribution Plot of β_8 in Bayesian Model 1

The 95% credible interval for *restecg* is [-0.129, 1.34], which includes zero, suggesting **weak posterior evidence** of a consistent association with the probability of heart disease.

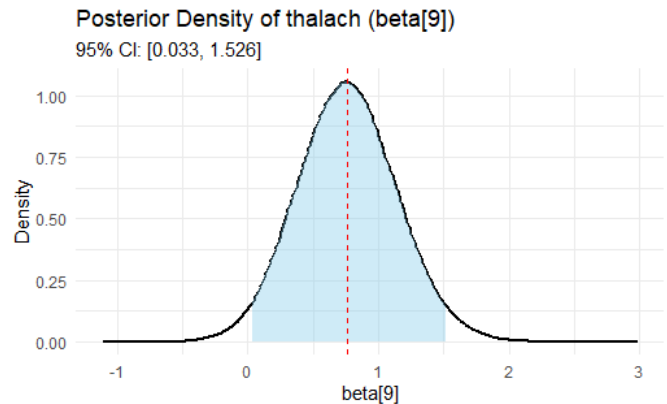


Fig. 11. Posterior Distribution Plot of β_9 in Bayesian Model 1

The posterior distribution of *thalach* lies entirely above zero, with a 95% credible interval [0.033, 1.526], indicating strong posterior evidence of a **positive association** with heart disease probability.

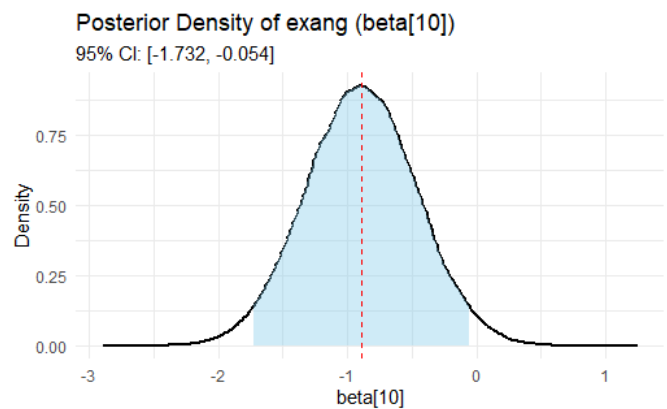


Fig. 12. Posterior Distribution Plot of β_{10} in Bayesian Model 1

The posterior distribution of *exang* lies entirely below zero, with a 95% credible interval [-1.732, -0.054]. This suggests strong posterior evidence of a **negative association** between exercise-induced angina and heart disease probability.

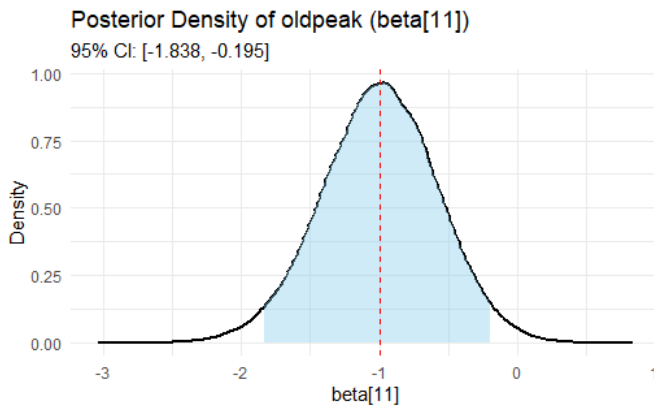


Fig. 13. Posterior Distribution Plot of β_{11} in Bayesian Model 1

The posterior distribution of *oldpeak* lies entirely below zero, with a 95% credible interval [-1.838, -0.195], indicating a **negative association** with the probability of heart disease.

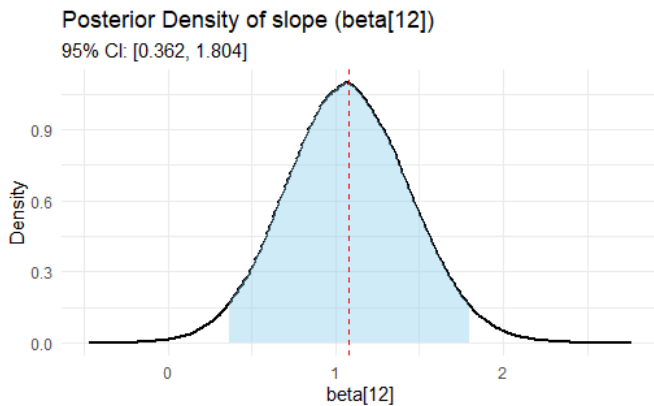


Fig. 14. Posterior Distribution Plot of β_{12} in Bayesian Model 1

The posterior distribution of *slope* lies entirely above zero, with a 95% credible interval [0.362, 1.804], indicating a **positive association** with heart disease probability.

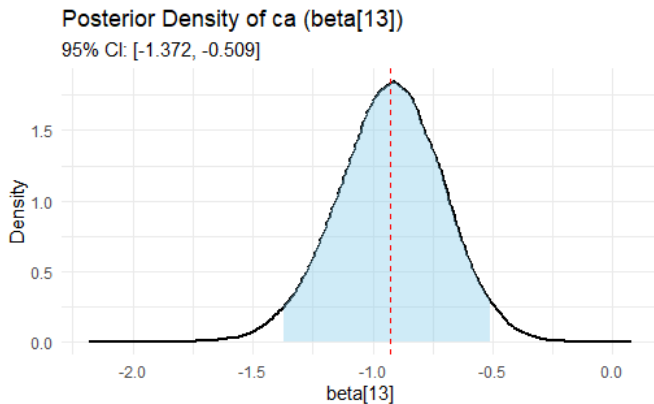


Fig. 15. Posterior Distribution Plot of β_{13} in Bayesian Model 1

The posterior distribution of *ca* lies entirely below zero, with a 95% credible interval [-1.372, -0.509], indicating a **negative association**, where a higher number of major vessels is associated with a lower probability of heart disease.

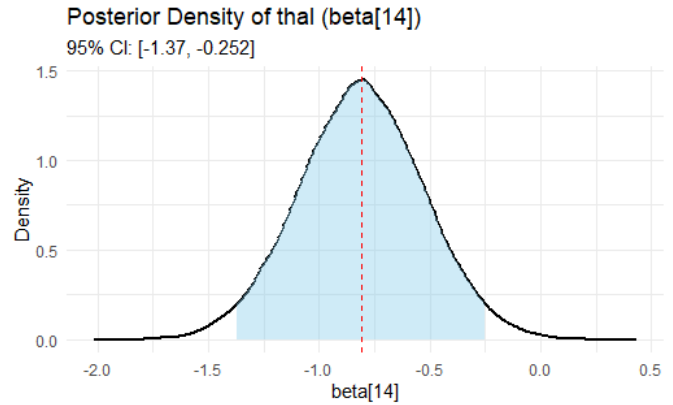


Fig. 16. Posterior Distribution Plot of β_{14} in Bayesian Model 1

The posterior distribution of *thal* lies entirely below zero, with a 95% credible interval [-1.37, -0.252], indicating strong posterior evidence of a **negative association** between *thal* and the presence of heart disease.

2. Bayesian Logistic Regression Model 2

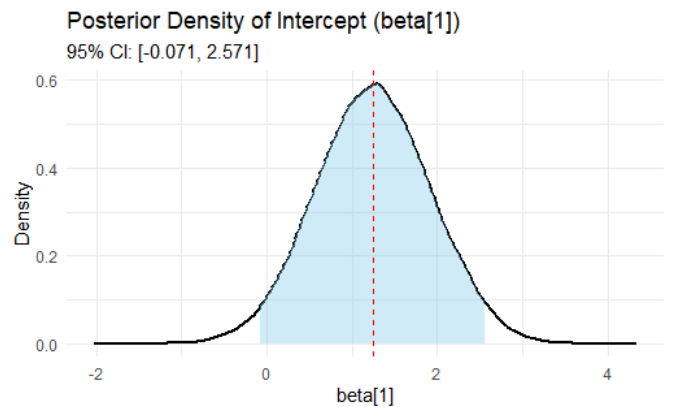


Fig. 17. Posterior Distribution Plot of β_1 in Bayesian Model 2

The intercept captures the baseline log-odds of heart disease when all predictor variables are held at their reference levels (categorical variables) or zero (numerical variables). Its posterior distribution indicates a moderately high but uncertain baseline risk.

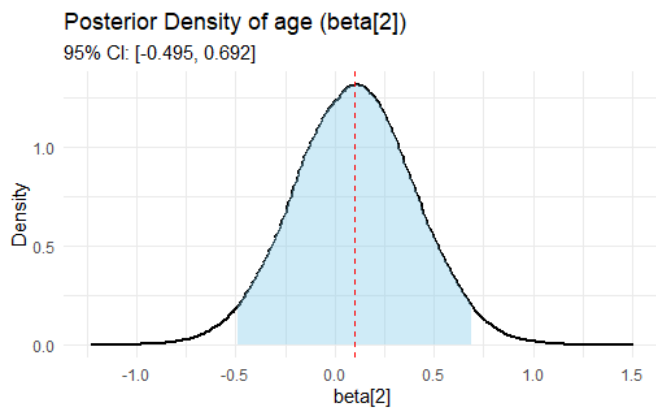


Fig. 18. Posterior Distribution Plot of β_2 in Bayesian Model 2

The 95% credible interval for *age* is $[-0.495, 0.692]$, which includes zero, suggesting **weak posterior evidence** of a consistent association with the probability of heart disease.

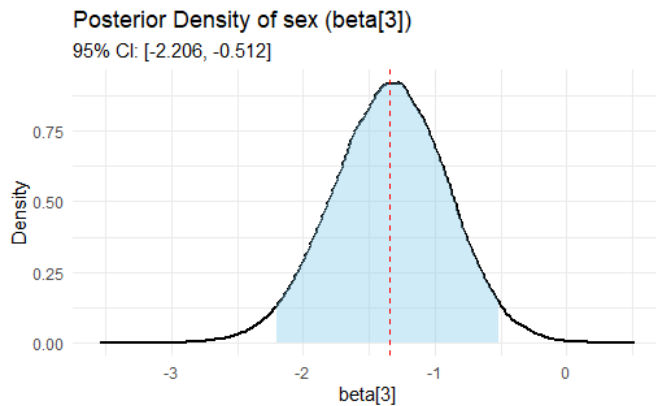


Fig. 19. Posterior Distribution Plot of β_3 in Bayesian Model 2

The posterior distribution of *sex* lies entirely below zero, with a 95% credible interval $[-2.206, -0.512]$, indicating a **negative association** with the probability of heart disease. This suggests that one sex category is associated with a lower probability of heart disease compared to the reference group.

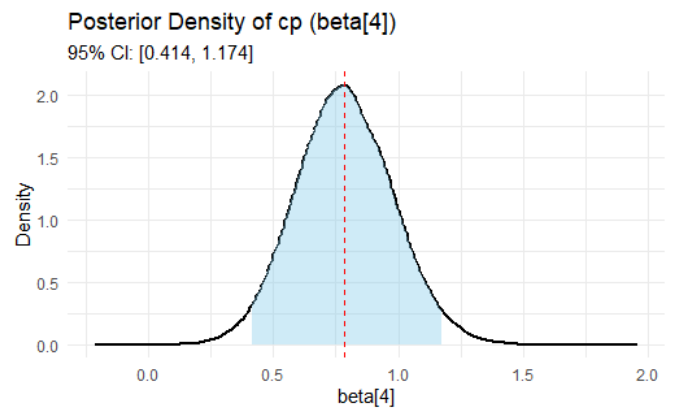


Fig. 20. Posterior Distribution Plot of β_4 in Bayesian Model 2

The posterior distribution of *cp* lies entirely above zero, with a 95% credible interval $[0.414, 1.174]$, indicating strong posterior evidence of a **positive association** with heart disease probability. This suggests that chest pain type is an important predictor of heart disease presence.

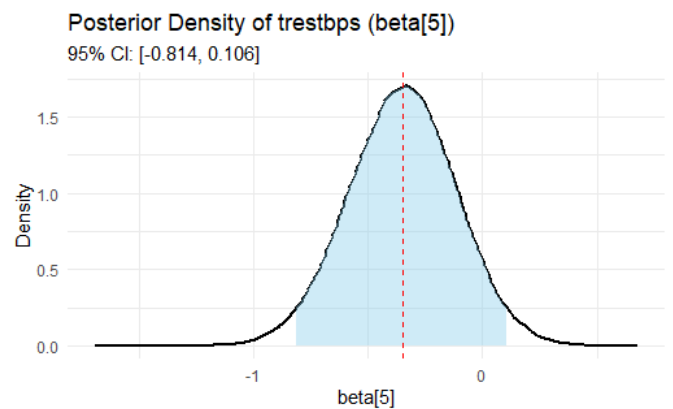


Fig. 21. Posterior Distribution Plot of β_5 in Bayesian Model 2

The posterior distribution of *trestbps* has a 95% credible interval $[-0.814, 0.106]$, which includes zero. This indicates **weak posterior evidence** of a systematic association between *trestbps* and the probability of heart disease.

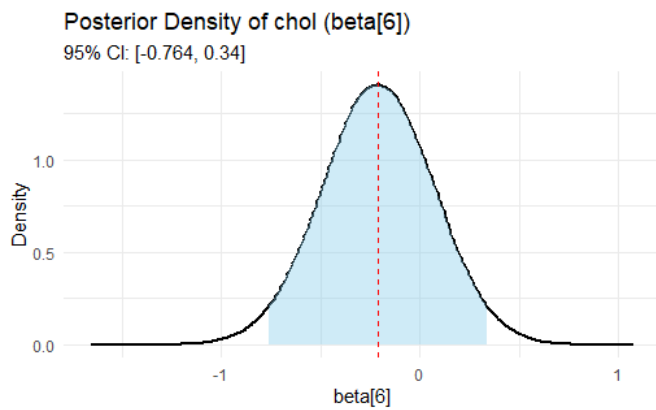


Fig. 22. Posterior Distribution Plot of β_6 in Bayesian Model 2

The posterior distribution of *chol* has a wide 95% credible interval [-0.764, 0.34], which includes zero, indicating **weak posterior evidence** of an association with heart disease.

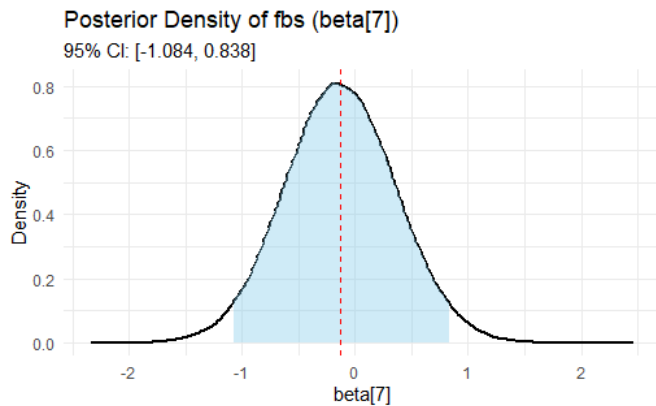


Fig. 23. Posterior Distribution Plot of β_7 in Bayesian Model 2

The posterior distribution of *fbs* has a wide 95% credible interval [-1.084, 0.838], which includes zero, indicating **weak posterior evidence** of an association with heart disease.

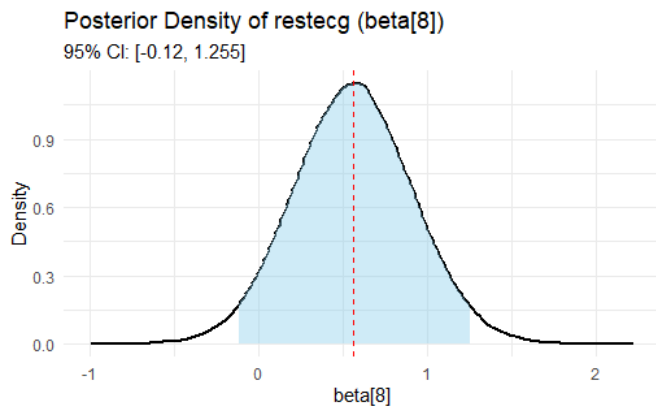


Fig. 24. Posterior Distribution Plot of β_8 in Bayesian Model 2

The 95% credible interval for *restecg* is [-0.12, 1.255], which includes zero, suggesting **weak posterior evidence** of a consistent association with the probability of heart disease.

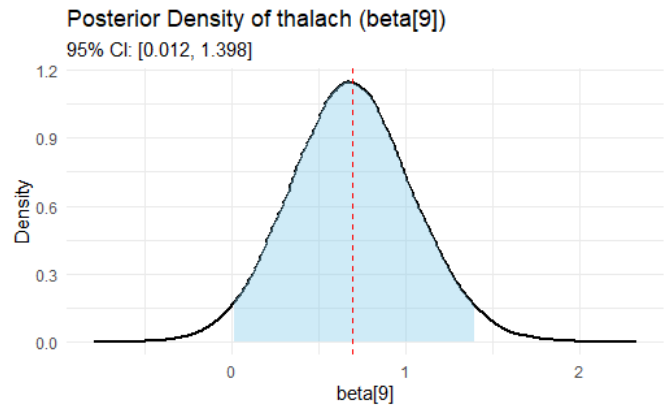


Fig. 25. Posterior Distribution Plot of β_9 in Bayesian Model 2

The posterior distribution of *thalach* lies entirely above zero, with a 95% credible interval [0.012, 1.398], indicating strong posterior evidence of a **positive association** with heart disease probability.

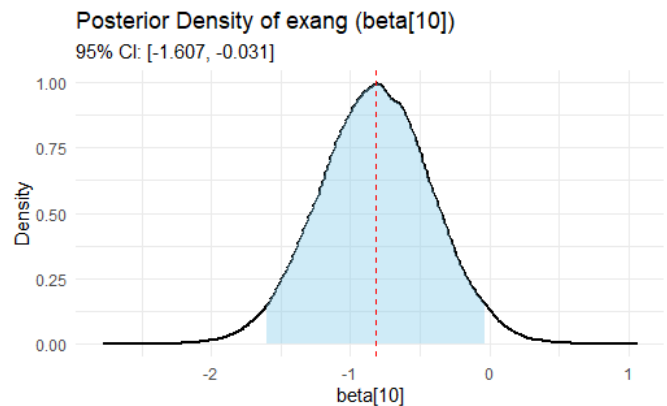


Fig. 26. Posterior Distribution Plot of β_{10} in Bayesian Model 2

The posterior distribution of *exang* lies entirely below zero, with a 95% credible interval [-1.607, -0.031], indicating a **negative association** with the probability of heart disease.

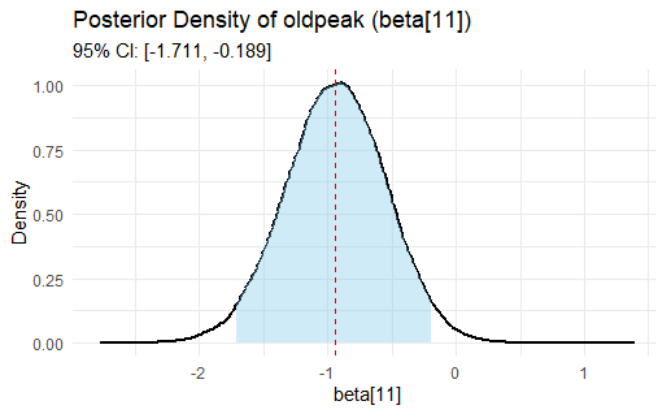


Fig. 27. Posterior Distribution Plot of β_{11} in Bayesian Model 2

The posterior distribution of *oldpeak* lies entirely below zero, with a 95% credible interval [-1.711, -0.189], indicating a **negative association** with the probability of heart disease.

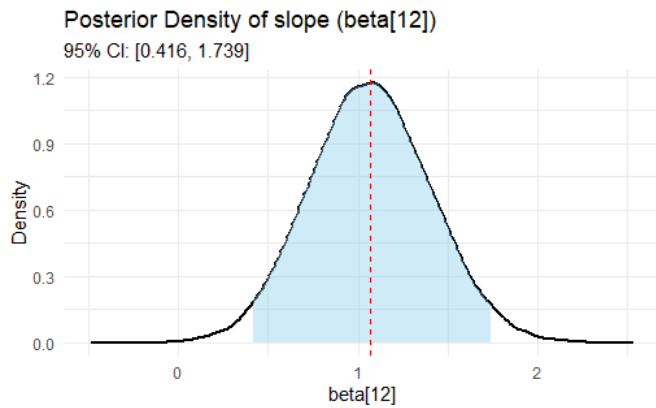


Fig. 28. Posterior Distribution Plot of β_{12} in Bayesian Model 2

The posterior distribution of *slope* lies entirely above zero, with a 95% credible interval [0.416, 1.739], indicating a **positive association** with heart disease probability.

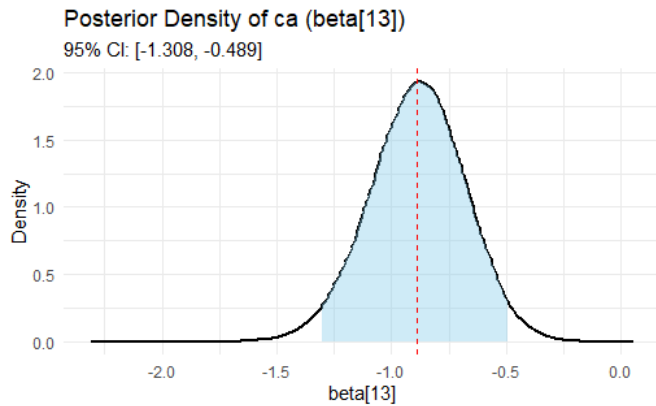


Fig. 29. Posterior Distribution Plot of β_{13} in Bayesian Model 2

The posterior distribution of *ca* lies entirely below zero, with a 95% credible interval [-1.308, -0.489], indicating a **negative association** with the probability of heart disease.

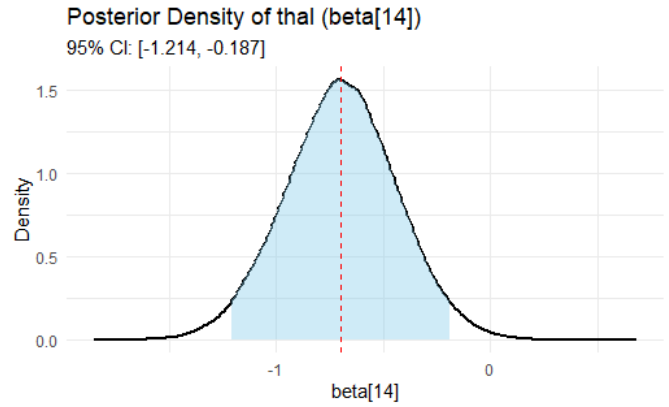


Fig. 30. Posterior Distribution Plot of β_{14} in Bayesian Model 2

The posterior distribution of *thal* lies entirely below zero, with a 95% credible interval [-1.214, -0.187], indicating strong posterior evidence of a **negative association** between *thal* and the presence of heart disease.

C. Model Evaluation

1. Bayesian Logistic Regression Model 1 Performance Metrics

Bayesian Logistic Regression Model 1	
Accuracy	0.8333
Precision	0.8286
Recall	0.8788
F1 Score	0.8529

Posterior Predictive Check

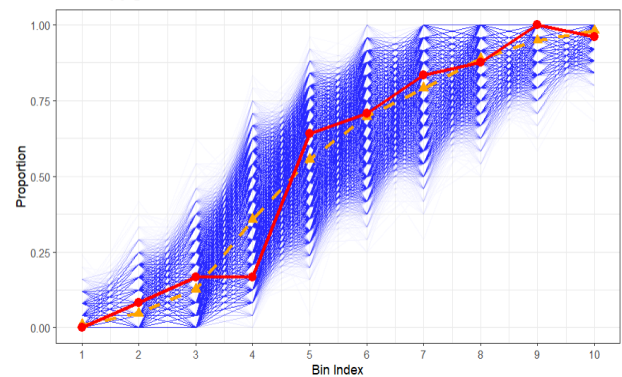


Fig. 31. PPC Plot Bayesian Model 1

The posterior predictive check shows that the observed proportions are largely contained within the posterior predictive distributions, indicating that the Bayesian logistic regression model provides an adequate fit to the data. The predictive samples closely follow the overall increasing trend of the observed values across the bins, suggesting that the model successfully captures the underlying relationship. Variability is higher in the lower bins, reflecting greater uncertainty at smaller outcome probabilities, while predictions become more stable at higher bins. Overall, the results indicate that the model with a normal prior (mean = 0, variance = 2) is well calibrated and capable of reproducing the observed data patterns.

ROC Curve

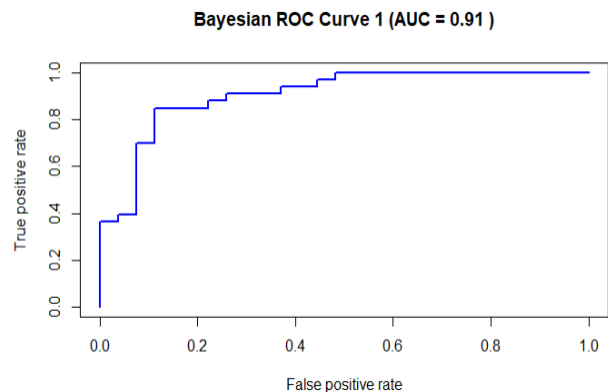


Fig. 32. ROC Plot Bayesian Model 1

The ROC curve indicates strong discriminative performance of the Bayesian logistic regression model. An AUC value of 0.91 suggests that the model has a high ability to distinguish between the positive and negative classes across different classification thresholds. The curve remains well above the diagonal reference line, indicating that the true positive rate increases rapidly with only a small increase in the false positive rate. Overall, these results demonstrate that the model provides accurate and reliable classification performance.

PR Curve

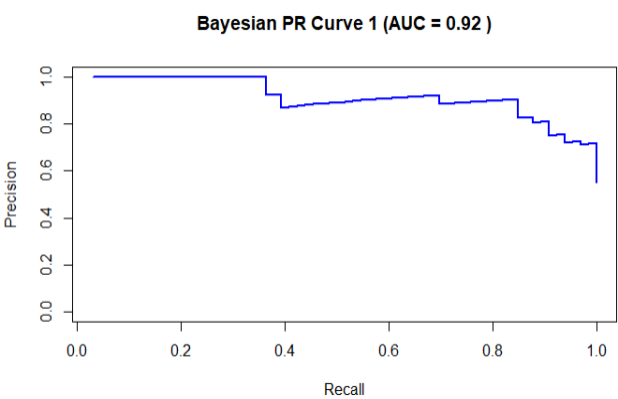


Fig. 33. PR Plot Bayesian Model 1

The precision recall curve demonstrates strong classification performance of the Bayesian logistic regression model, particularly in identifying the positive class. An AUC value of 0.92 indicates that the model maintains high precision across a wide range of recall levels. The curve shows that precision remains relatively high even as recall increases, suggesting that the model is effective at correctly identifying positive cases while limiting false positives. Overall, these results indicate reliable predictive performance, especially in scenarios where class imbalance may be present.

Trace Plot

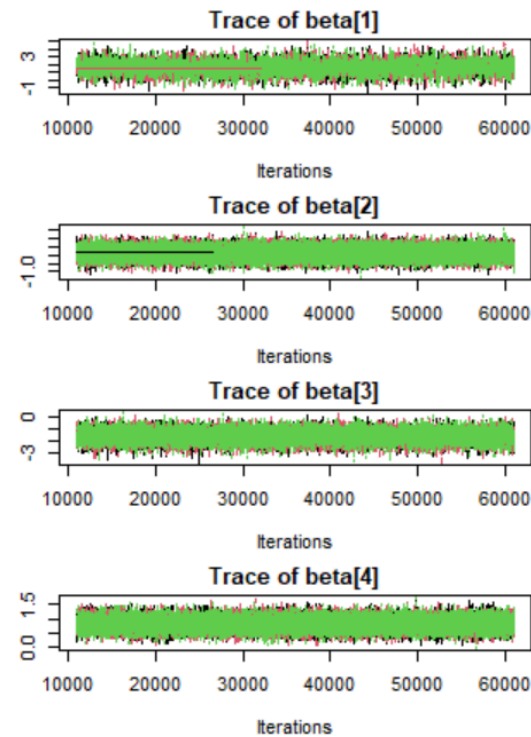


Fig. 34. Trace Plot Bayesian Model 1 ($\beta_1 - \beta_4$)

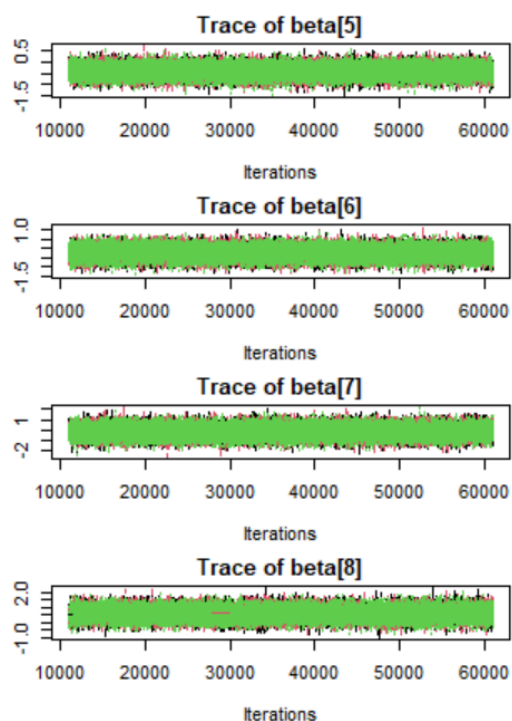


Fig. 35. Trace Plot Bayesian Model 1 ($\beta_5 - \beta_8$)

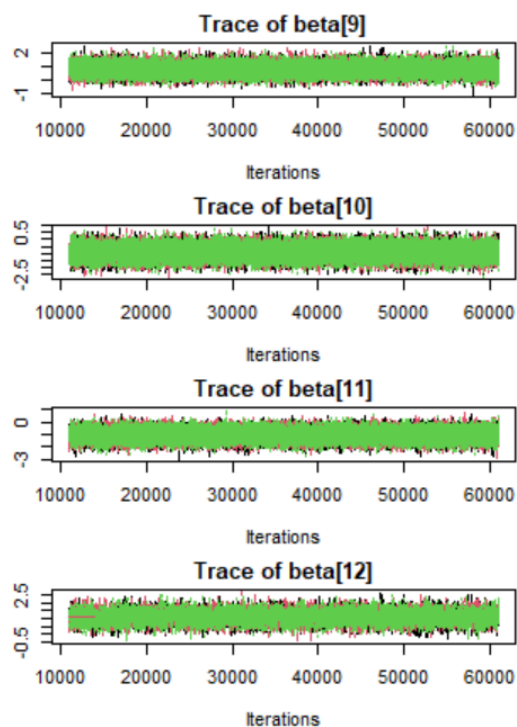


Fig. 36. Trace Plot Bayesian Model 1 ($\beta_9 - \beta_{12}$)

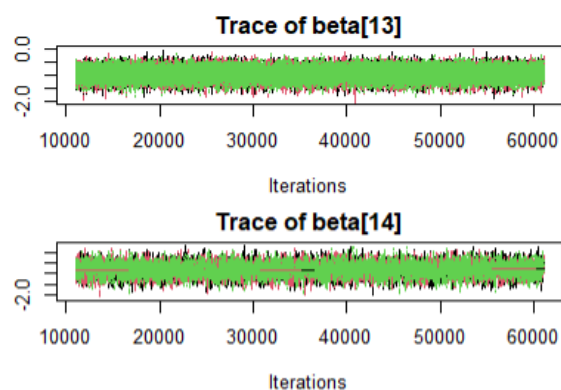


Fig. 37. Trace Plot Bayesian Model 1 ($\beta_{13} - \beta_{14}$)

The trace plots for the Bayesian model with variance set to 2, based on 50000 MCMC iterations, indicate good convergence behavior. The parameter samples fluctuate around stable mean values throughout the iterations, with no visible trends, systematic drifts, or abrupt jumps. The chains appear well mixed and consistently explore the posterior distribution across the entire sampling process. These patterns suggest that the MCMC algorithm has successfully converged, and the resulting posterior estimates can be considered reliable for further inference and model interpretation.

Gelman-Rubin Statistic

Potential scale reduction factors:

	Point est.	Upper C.I.
beta[1]	1	1
beta[2]	1	1
beta[3]	1	1
beta[4]	1	1
beta[5]	1	1
beta[6]	1	1
beta[7]	1	1
beta[8]	1	1
beta[9]	1	1
beta[10]	1	1
beta[11]	1	1
beta[12]	1	1
beta[13]	1	1
beta[14]	1	1

Multivariate psrf

1

The gelman rubin convergence diagnostic (potential scale reduction factor, PSRF) indicates that all regression parameters (β_1 to β_{14}) have point estimates equal to 1 with upper confidence intervals also equal to 1. In addition, the multivariate PSRF value is 1.

PSRF values equal to or very close to 1 suggest that the between-chain and within-chain variances are well

balanced, indicating strong agreement among the MCMC chains. This result provides clear evidence that the Markov Chain Monte Carlo sampling process has successfully converged for all parameters. Consequently, the posterior distributions can be considered stable, and the resulting posterior summaries, such as posterior means and credible intervals, are reliable for subsequent inference and analysis.

2. Bayesian Logistic Regression Model 2
Performance Metrics

Bayesian Logistic Regression Model 2	
Accuracy	0.8500
Precision	0.8529
Recall	0.8788
F1 Score	0.8657

Posterior Predictive Check

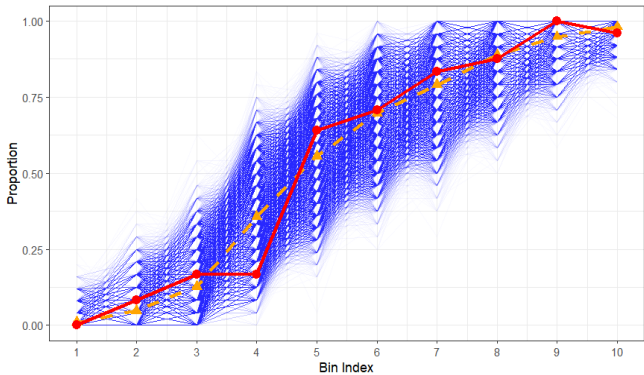


Fig. 38. PPC Plot Bayesian Model 2

The posterior predictive check indicates that the observed proportions remain largely within the posterior predictive distributions, showing that the model still provides an adequate fit. Compared to the larger-variance prior, the predictive distributions are more concentrated, reflecting reduced uncertainty due to stronger prior shrinkage. The model continues to capture the increasing trend across bins, although minor deviations are visible in the lower bins. Overall, the Bayesian logistic regression with a normal prior (mean = 0, variance = 1) yields more conservative and stable predictions while maintaining reasonable agreement with the observed data.

ROC Curve

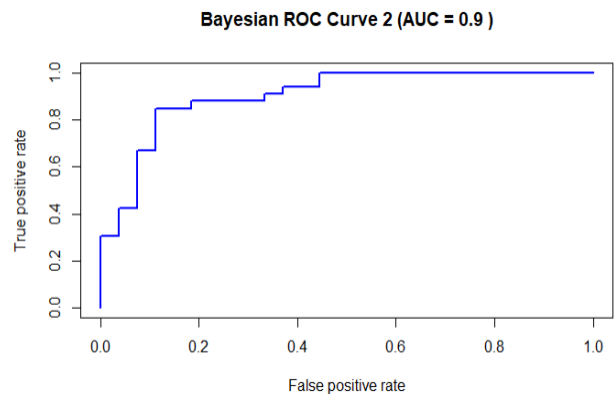


Fig .39. ROC Plot Bayesian Model 2

The ROC curve indicates that the Bayesian logistic regression model exhibits strong discriminative capability. An AUC value of 0.9 suggests that the model is highly effective in distinguishing between positive and negative classes across different classification thresholds. The curve lies well above the diagonal reference line, indicating a high true positive rate achieved with relatively low false positive rates. Overall, the results demonstrate reliable and robust classification performance.

PR Curve

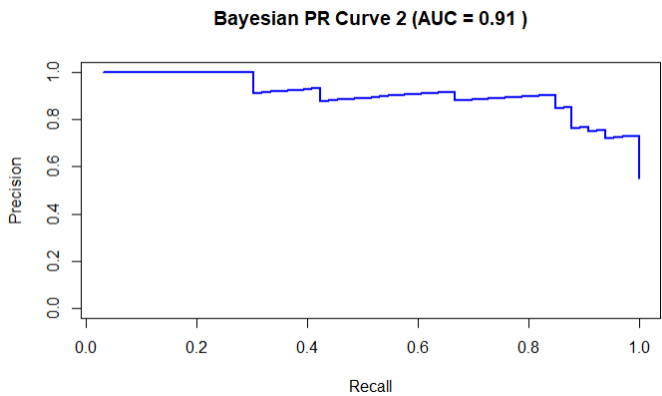


Fig. 40. PR Plot Bayesian Model 2

The precision recall curve indicates strong predictive performance of the Bayesian logistic regression model in identifying the positive class. An AUC value of 0.91 suggests that the model maintains high precision across a broad range of recall values. The curve shows that precision remains relatively stable as recall increases, indicating effective detection of positive instances while controlling false positives. Overall, the results demonstrate reliable and well-calibrated classification performance.

Trace Plot

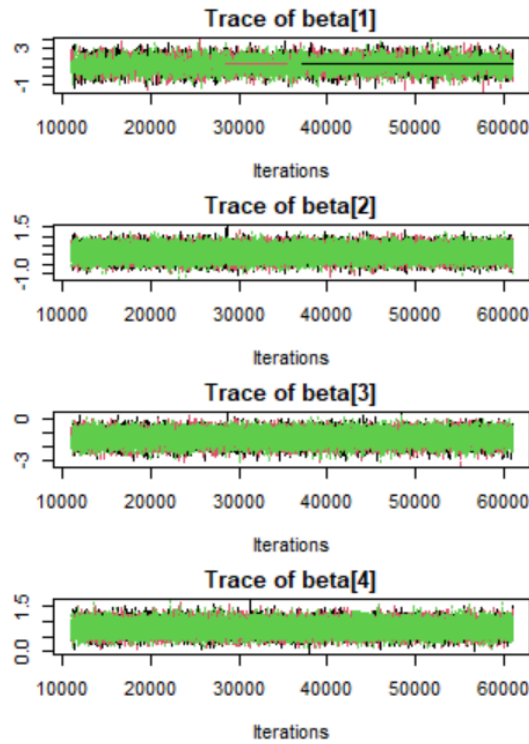


Fig. 41. Trace Plot Bayesian Model 2 ($\beta_1 - \beta_4$)

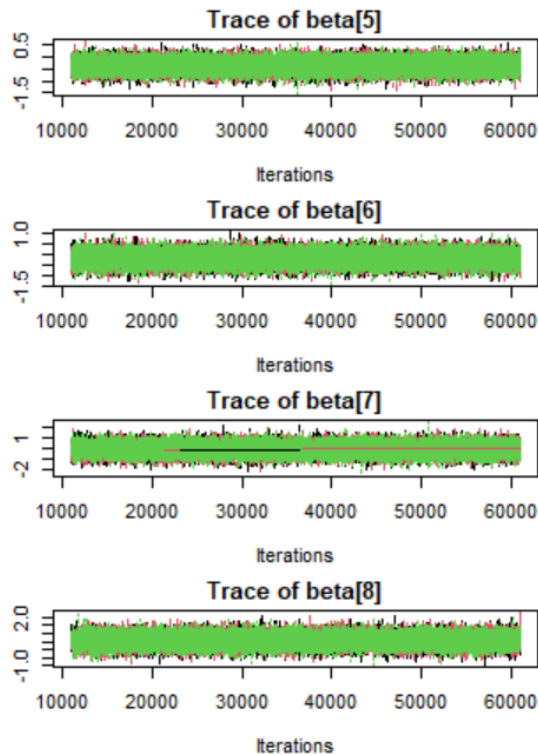


Fig. 42. Trace Plot Bayesian Model 2 ($\beta_5 - \beta_8$)

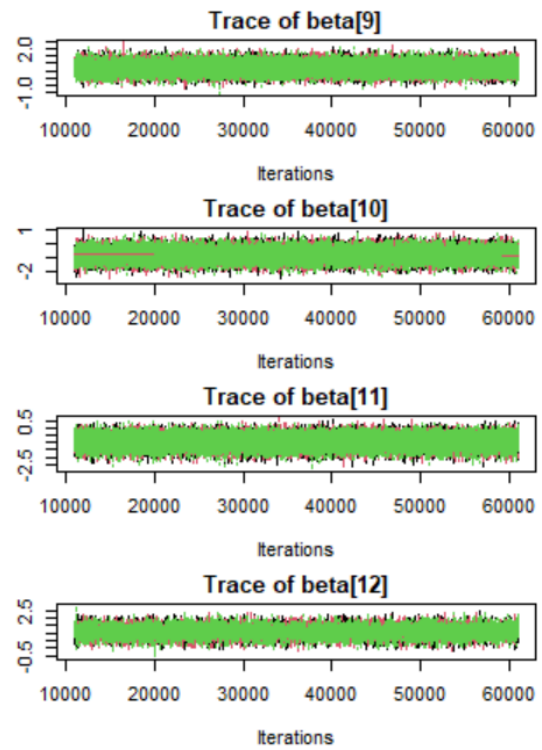


Fig. 43. Trace Plot Bayesian Model 2 ($\beta_9 - \beta_{12}$)

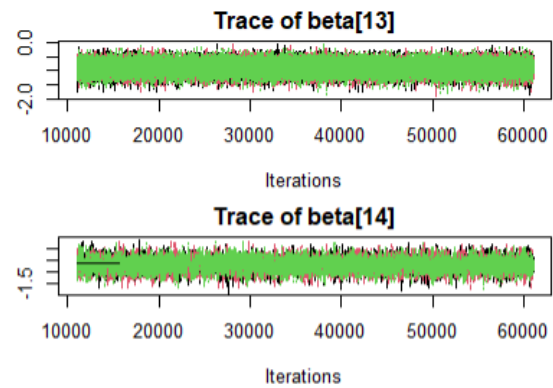


Fig. 44. Trace Plot Bayesian Model 2 ($\beta_{13} - \beta_{14}$)

The trace plots for the Bayesian model with variance set to 1, based on 50000 MCMC iterations, indicate good convergence behavior. The parameter samples fluctuate around stable mean values throughout the iterations, with no visible trends, systematic drifts, or abrupt jumps. The chains appear well mixed and consistently explore the posterior distribution across the entire sampling process. These patterns suggest that the MCMC algorithm has successfully converged, and the resulting posterior estimates can be considered reliable for further inference and model interpretation.

Gelman-Rubin Statistic

Potential scale reduction factors:

	Point est.	Upper C.I.
beta[1]	1	1.01
beta[2]	1	1.00
beta[3]	1	1.00
beta[4]	1	1.00
beta[5]	1	1.00
beta[6]	1	1.00
beta[7]	1	1.00
beta[8]	1	1.00
beta[9]	1	1.00
beta[10]	1	1.00
beta[11]	1	1.00
beta[12]	1	1.00
beta[13]	1	1.00
beta[14]	1	1.00

Multivariate psrf

1

The gelman rubin convergence diagnostic, summarized by the potential scale reduction factor (PSRF), indicates satisfactory convergence for all regression coefficients (β_1 to β_{14}). The point estimates of the PSRF are equal to 1 for all parameters, while the upper confidence interval values are equal to 1.00 for most coefficients and 1.01 for β_1 .

PSRF values close to 1 suggest that the within-chain variance is comparable to the between-chain variance, implying good mixing across Markov chains. The slightly higher upper confidence bound for β_1 remains well below commonly accepted convergence thresholds, indicating no practical convergence issues. Furthermore, the multivariate PSRF value of 1 confirms joint convergence of the parameter vector. These results demonstrate that the MCMC sampling process has converged adequately, and the posterior estimates are stable and suitable for further inference.

3. Frequentist

Performance Metrics

Frequentist	
Accuracy	0.8167
Precision	0.8235
Recall	0.8485
F1 Score	0.8358

ROC Curve

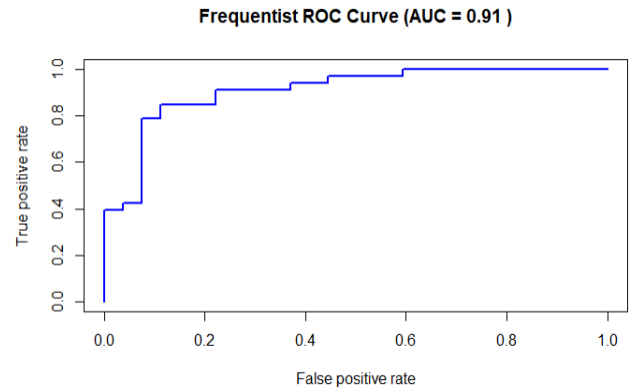


Fig. 45. ROC Plot Frequentist

The ROC curve demonstrates strong classification performance of the frequentist logistic regression model. An AUC value of 0.91 indicates a high ability of the model to distinguish between positive and negative classes. The curve shows a rapid increase in the true positive rate at relatively low false positive rates, suggesting effective sensitivity while maintaining control over false alarms. Overall, the model exhibits reliable discriminative capability across different decision thresholds.

PR Curve

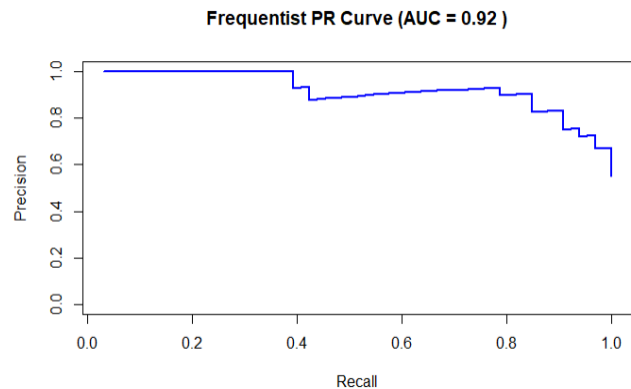


Fig. 46. PR Plot Frequentist

The precision recall curve indicates strong performance in identifying the positive class. An AUC value of 0.92 suggests that the model maintains high precision across a wide range of recall levels. Precision remains relatively stable at lower to moderate recall and gradually decreases as recall approaches its maximum, reflecting the typical trade-off between capturing more positive cases and increasing false positives. Overall, the model shows robust effectiveness in handling class prediction, particularly in scenarios where precision is an important consideration.

E. Model Comparison

Model	Bayesian Model 1 [Normal(0, 0.5)]	Bayesian Model 2 [Normal(0, 1)]
DIC	193.7	193.5
LOOIC	195.80	194.57
WAIC	195.54	194.38

The comparison of the two Bayesian models shows consistent results across all Bayesian information criteria. Bayesian Model 2 exhibits slightly lower DIC (193.5) compared to Bayesian Model 1 (193.7), indicating a better trade off between model fit and complexity. A similar pattern is observed for LOOIC, where Bayesian Model 2 achieves a lower value (194.57) than Model 1 (195.80), suggesting superior out-of-sample predictive performance and generalization ability. Similarly, in terms of WAIC, Bayesian Model 2 produces a smaller value (194.38) compared to Bayesian Model 1 (195.54), further confirming the advantage of Model 2 in posterior based predictive accuracy. Although the numerical differences are relatively small, the consistent improvement across DIC, LOOIC, and WAIC provides strong evidence that Bayesian Model 2 with a Normal(0, 1) prior offers greater model flexibility and is the preferred model for this study.

Model	Bayesian Model 1 [Normal(0, 0.5)]	Bayesian Model 2 [Normal(0, 1)]	Frequentist
Accuracy	0.8333	0.8500	0.8167
Precision	0.8286	0.8529	0.8235
Recall	0.8788	0.8788	0.8485
F1-Score	0.8529	0.8657	0.8358
ROC	0.9057	0.9035	0.9091
PR	0.9217	0.9152	0.9108

The comparison across all models indicates that Bayesian Model 2 with a Normal(0, 1) prior provides the best overall predictive performance. Bayesian Model 2 achieves the highest accuracy (0.85), precision (0.8529), and F1-score (0.8657), indicating a better balance between correctly identifying positive cases and minimizing misclassification errors. Both Bayesian models exhibit higher recall (0.8788) compared to the frequentist model (0.8485) suggesting greater

sensitivity in detecting heart disease cases. Although the frequentist model slightly outperforms the Bayesian models in terms of ROC-AUC (0.9091), the difference is relatively small. In terms of the Precision-Recall metric, Bayesian Model 1 records the highest value (0.9217). However, Bayesian Model 2 remains competitive (0.9152) and demonstrates more stable performance across most evaluation metrics. Overall, these results confirm that Bayesian Model 2 is the most well balanced and superior model among all evaluated approaches.

IV. CONCLUSION

A. Summary of findings

This study demonstrates that model performance in predicting heart disease is strongly influenced by both the inferential framework and prior specification. Among the Bayesian approaches, Bayesian Model 2 with a Normal(0, 1) prior consistently outperformed Bayesian Model 1, as indicated by lower DIC, LOOIC, and WAIC values, reflecting a superior balance between model fit, complexity, and predictive generalization. In terms of predictive accuracy, Bayesian Model 2 achieved the highest accuracy, precision, and F1-score, indicating the most balanced classification performance. Both Bayesian models exhibited higher recall than the frequentist model, suggesting improved sensitivity in detecting heart disease cases. Although the frequentist logistic regression achieved the highest ROC-AUC, the difference was marginal and did not translate into superior overall predictive balance. Taken together, these findings indicate that Bayesian logistic regression with moderately informative priors provides the most robust and reliable predictive framework for this dataset, offering clear advantages over the classical frequentist approach in terms of uncertainty handling and decision-oriented performance.

B. Limitations

This study has several limitations. The dataset used has a limited sample size and may not fully represent diverse clinical populations, which can restrict the generalizability of the results. The analysis is also confined to logistic regression models, and therefore does not capture potentially complex nonlinear relationships among predictors. Furthermore, Bayesian inference is performed using Markov Chain Monte Carlo (MCMC), which is computationally more intensive than frequentist estimation and may be sensitive to sampling settings such as the number of iterations, burn in period, and convergence diagnostics. In addition, the prior specifications examined in this study represent only a subset of possible choices, and alternative priors may lead to different posterior inferences and predictive performance.

C. Future Work

Future studies may address these limitations by employing larger and more diverse heart disease datasets to enhance generalizability across clinical populations. The comparison can be extended beyond logistic regression to include more flexible models, such as Bayesian hierarchical or nonlinear models, to better capture complex predictor outcome relationships. Moreover, alternative Bayesian inference methods, including Hamiltonian Monte Carlo or variational inference, could be explored to improve computational efficiency and scalability. Finally, future research should consider a broader range of prior specifications and conduct systematic sensitivity analyses to more comprehensively assess the impact of prior choice on posterior inference and predictive performance.

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